

Table 2. Recommendation adherence

Table 2: Recommendation adherence by Brown et al. recommendation window and day type. Observed valid-minute fractions are descriptive; work-day and free-day estimates and their difference are model-based, with 95% confidence intervals and false-discovery-rate-adjusted p-values. Each of nine sites receives equal weight in the model estimates. Daytime, Pre-sleep and Sleep identify the recommendation windows: daytime excludes the three hours before reported sleep, pre-sleep comprises those three hours, and sleep describes the bedside sleep environment. The wake-start date assigns the day type to the preceding sleep, daytime and following pre-sleep windows; each window qualifies independently. Only the primary any-valid sample is shown: 2,298 periods, 140 participants, 794 cycles and 1,043,192 valid minutes. The 80% coverage sensitivity remains reported in the Results and Methods; its pre-sleep interval includes zero, and temporal dependence remains unresolved. Confidence-interval exclusion and FDR decisions are separate summaries. Descriptive pooled-minute fractions are detailed in Supplementary Table S3. Adherence does not classify participants.

Recommendation adherence by recommendation window and day type						
Recommendation window	Recommendation	Observed pooled-minute adherence	Site-average recommendation-window model			
		Valid minutes meeting recommendation, n/N (%)	Work-day adherence % (95% CI)	Free-day adherence % (95% CI)	Free minus Work percentage points (95% CI)	FDR-adjusted p
Daytime	At least 250 lx melanopic EDI during wake, excluding the three hours before sleep	137,792/573,712 (24.0%)	26.6 (24.5 to 28.7)	21.5 (19.2 to 23.8)	-5.1 (-7.8 to -2.4)	<0.001
Pre-sleep	No more than 10 lx melanopic EDI during the three hours before reported sleep	81,894/129,390 (63.3%)	62.0 (58.8 to 65.2)	67.9 (64.2 to 71.6)	+5.9 (+1.4 to +10.5)	0.011
Sleep	No more than 1 lx melanopic EDI during reported sleep; bedside sleep environment	336,052/383,366 (87.7%)	90.4 (88.9 to 91.9)	83.9 (81.4 to 86.5)	-6.5 (-9.1 to -3.8)	<0.001

Observed pooled-minute fractions and fitted recommendation-window estimates are different quantities. Each of the nine sites contributes equally to the fitted estimates; the displayed model results use only the primary any-valid sample.

The recommendation boundaries are inclusive: daytime ≥ 250 lx, pre-sleep ≤ 10 lx and sleep ≤ 1 lx melanopic EDI. Near-eye measurements describe ocular exposure during wear; reported sleep describes the bedside sleep environment, not worn ocular exposure. Chest measurements are not pooled here.

Day type is assigned from the wake-start date to the preceding sleep, daytime and following pre-sleep windows; windows qualify independently. Temporal dependence remained unresolved. The 80% coverage sensitivity, including the inconclusive pre-sleep interval, is reported in the Results and Methods, not as additional rows here.